

Application of the Free Energy Principle to Estimation and Control

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[Music] welcome to this presentation on the application of the free energy principle to estimation and control my name is Hank ramesh nama Chalmers University of Technology this is joint work with test from the lag from tu and Hoeven and our cha or Chile Cali from Uppsala University the motivation for this work lies in the area of stochastic optimal control the picture on the right shows an agent which can act on its environment in first the state of the environment through sensors and observations and collects rewards using a control theoretical tation we have the following ingredients the state X which can contain the state of the agent and the world the control or action u the observations Y and finally the goals a classic example is LQG which stands for linear quadratic Gaussian control here the agent has linear dynamics and a quadratic cost quadratic in both the state and the control LQG features a so called separation principle where the control assumes a point estimate of the state and ignores any uncertainty of the state and on the other hand the inference is done by a common filter which assumes known past controls this decouples the control and inference problem in more general context inference can be solved through the belief propagation algorithm of which the Kalman filter is a specific instance in addition to this classic example there exist more modern approaches such as model predictive control and reinforcement learning a model predictive control we assume to have knowledge of a generative model of the agent dynamics as well as the observation model the goals are encoded with a loss or objective function and based on this the model predictive controller solves an optimization problem to determine the current control action on the other hand reinforcement learning typically is model-free so it does not have a generative model and it builds value functions or policies the goals of the agent are recorded via a reward signal in this work we will use a third way called active inference based on the reference below roughly speaking active inference tries to minimize free energy or entropy or taking the quote from this paper it acts to minimize surprise avoiding surprise is a

fundamental for survival and speaks to the basic need of organisms to maintain equilibrium with their environment the main question that we all want to answer in this work is the following can we express stochastic optimal control in general and LQG control specifically as special cases of active inference where this motivation in mind this brings me to the outline of this presentation first I will provide some brief background on LQG and belief propagation then I will elaborate on different ways how a belief propagation has been used for control problems then we move on to the main topic which is the use of active inference for control and a relation to LQG I will then show one example result and wrap up with conclusions so let's first start with LQG and belief propagation let us start with LQG linear quadratic Gaussian control I have taken this information on this slide from Wikipedia the first box shows the evolution of the state as well as the observation model the state at $t + 1$ is a linear function of the state at time t and the control perturbed by Gaussian noise the observation at time t is a linear function of the state at time t plus independent Gaussian noise so this is the linear part of the LQG the quadratic part refers to the cost the cost to be minimized is a quadratic function in the state and in the control there's an expectation over all random variables so basically we want our agent to stay close to zero and do this with as low control as possible it turns out that this system can be solved in closed form and the discrete time LQG controller is given by these equations the first equation gives us the evolution of the state given the control and this is basically the equation from the Kalman filter the estimate of the state at time t so here the $\hat{x}$ refers to an estimate depends on the previous estimate the control and a correction factor which depends on the current observation give it an estimate of the state we can then find a control by a linear feedback so we have a linear function depends on the state estimate now in these expressions there are several matrices and this can be found from the so called Riccati equations solving the Riccati equations provides the matrices needed to find the discrete-time LQG controller the figure on the right visualizes the LQG controller so we have a Kalman filter which takes into account this control input the observation generates a state estimate then there's a multiplication with minus K which yields the control the control is applied to the plant or the system and then this generates via a sensor and new observation which again is fed to the Kalman filter we now move on to belief propagation in belief propagation we consider distributions over multiple variables such as here $P(x_1, x_2, x_3)$ we assume that this distribution can be factorized as follows so there's a normalization constant $1/Z$ and then 3 factors F_A, F_B and F_C where each factor contains a subset of the arguments of the original distribution from this factorization we can draw a factor graph in this

factor graph each vertex for each node correspond to a factor and each edge or line corresponds to a variable so for instance a corresponds to F_A X_1 corresponds to the variable X_1 we see that a only connects to X_1 so this means that F_A has only one variable namely X_1 but X_1 connects to a and B this is because X_1 appears in F_A and F_B what's this factor graph we can run the belief propagation algorithm this is a message passing algorithm over the factor graph the messages we pass are functions of The Associated variables so again we here have the factor graph and it will now pass messages from this from the edges of the fact graph in word so we first compute a message from a to X_1 which we denote by μ from F_A to X_1 evaluate it and the value X_1 and this message or this function is just given by the original function F_A of X_1 at the same time we can compute a message from X_3 to see some you from X_3 to F_C evaluated in the value X_3 I will just set this to 1 now given these two messages we can compute messages from F_B to X_2 and F_C to X_2 both are functions of the variable X_2 the message from F_B to X_2 evaluated the value X_2 is given by the function F_B multiplied by the incoming messages so that's the message from a to X_1 and then we integrate out all the variables except X_2 similarly the message from F_C to X_2 is the function F_C which depends on X_2 and X_3 multiplied by the incoming message which we've computed in the previous time step and that we integrate out over everything except X_3 finally we can compute these messages from F_B to X_1 and from F_C to X_3 following the same rules after this third time step on each edge we have two messages one from left to right and one from right to left given these messages we can compute so-called beliefs the belief of x_1 the belief of x_2 and the belief of x_3 and these are simply the marginals of the original distribution so $p(x_1)$ is obtained by taking the original distribution and integrating out everything except x_1 and this belief which is equal to the marginal is given by the product of the two messages the message from left to right from a to x_1 and a message from right to left from B to x_1 the same holds for x_2 and x_3 and this provides a systematic and efficient way to infer hidden variables it can be shown that these beliefs are exact when the graph has no cycles when the factor graph has cycles or loops the beliefs are only approximations belief propagation has many applications including the Kalman filter the particle filter decoding of error correcting codes and image processing to name but a few we now focus on the relation between belief propagation and free energy minimization because an active inference we also minimize the type of free energy to reveal this connection let's express the Joint Distribution P of X in a slightly different way so here X refers to the vector of all the variables so X can be a very high dimensional variable we write P of X as $1/Z$ and Z is a normalization constant and then a product over

functions so product a over possibly many functions F_A where each F_A takes as argument X_A where X_A is a subset of the original X we now introduce a new notation Y of X defined as the sum of the logarithm of the individual F_A s with this new notation Y of X we can express P of X as 1 over Z times the exponential of minus G of X here Y of X can be interpreted as a kind of energy so U of X tells us energy of a certain configuration X the distribution P of X will have most of its mass in low-energy configurations hence and free energy minimization the normalization constant Z can in principle be found by solving this integral ok now with this in mind let us now try to formulate a new optimization problem so we want to minimize with respect to Q so here Q is the set of distributions the divergence between Q and P and P is the original distribution block plugging in the definition of the Kullback-Leibler divergence we can express the divergence as an expected energy $Mises$ - an entropy minus $\log$ of Z so the Kullback-Leibler divergence has an average energy and entropy and then finally $\log$ of Z which is called a log partition function the first two terms are called the beta free energy since we want to minimize the divergence we try to find a Q that minimizes the average free energy the average energy and at the same time has high entropy so it's as broad as possible while minimizing the average energy now this is an optimization problem so we can in principle formulate it at the Lagrangian and solve this optimization problem and it turns out and this was shown in reference to that solving for Q of X given some structure on Q of X leads to the belief propagation message passing rules so this has this consequence that's when you're doing belief propagation you're actually solving a better free energy minimization problem namely this one with this background in mind we cannot move on to the relation between belief propagation and control let us assume that we have a generative model and we want to infer hidden States so what we can do is we can determine the Joint Distribution of the States and the observations given the controls due to the Markov assumption this distribution factorizes as the prior times the dynamic model and the observation likelihood let's further assume that our control objective is quadratic in the state and quadratic in the control now at time T our factor graph will look something like this we have a part of the factor graph that relates to the past where we have the past controls which are known because we've already decided them and also the observations up to time T which are also known because our sensor has observed them and then we have the future so this here contains the future controls which we do not know and the future observations which have not been observed yet the factor graph late to the past contains the states up to time T the factor graph of the future contains the states from time T onward what connects them here is the state at time T now on the factor graph

related to the past the one in green we can run belief propagation and based on this belief propagation we can find marginals or beliefs and these are of the form for instance of the distribution of the state at some time T prior to the current time T and including time T given the observations up to time T and all the past controls this can be found by just simply running belief propagation on this left side so to be a little bit more specific in the past we have messages with known controls and known observations so here are some arrows that represent these messages and let us focus on some of these at time T so the message here labeled a turns out to be the prediction so this is the distribution of the state at time T given the observations in the past so before time T and also the past controls so this is what we know about the state at time T before observing the measurement Y_T the message B is the likelihood so depends on the observation and then a message see which connects the past in the future is the update so this is the distribution of this data time T given all the measurements including Y_T and the past controls and this update of course following the message passing rules is the product the likelihood function and the prediction and these messages may seem very familiar to you because they are just the standard equations used in the Kalman filter and the particle filter particle filter now for the future even though we have this factor graph it is meaningless to do belief propagation the reason is that a control objective C_T is not encoded anywhere in this factor graph so there's no way to determine reasonable controls the way this problem has been addressed in the literature is by encoding the control objective directly in the factor graph so here again is our control stage cost which is quadratic in the states and quadratic in the control so what paper number one proposes is to create an artificial prior on future controls and this prior is proportional to an exponential negative the control cost so what this means for instance when H is very large is that our generative model tells us that we will end up in states with low control similarly to deal with stage costs related to the state we introduce artificial known observations and an associated likelihood function which is proportional to exponential negative state cost so again when R is very large this tells us that our generative model brings us to states where X is small we also remove future observations because those are not known this then brings us to the following factorization of the states from time 0 to time capital T observations up to time little T because future observations are removed the controls from time 1 to time capital T and then the artificial future observations from time T plus 1 come time capital T we express this using the standard Markov model as a partner to the past with the prior the state dynamics and the observation model and then a part related to the pewte to the future where we again have the stage state dynamics we have the prior over future controls

and the likelihood function related to future States here we show again the factor graph from the previous slide where the green part relates to the past the red part relates to the future and X of T Co connects the past with the future recall again in this graph the future controls and the future observations are unknown we now replace this part of the future with this factor graph on the right corresponding to the factorization above so what we did is for each control we added a prior on the control of the form described above for each state in the future we add a likelihood function with an Associated artificial observation equal to one now we also removed all the future observations Y_T so now X and u are unknown and Z is known but this is sufficient for us to run the belief propagation algorithm so we can do for message-passing where we start from the updated belief of X_T given all observations up to time little T and all controls up to time t minus 1 we pass messages to the future we pass messages from the future to the current time U_T now we have two messages on this edge U_T and we can compute the belief so following the belief propagation rule the belief U_T provides us the belief over the next control action we can take the mode of this belief and implement that action and then go through the entire process again now it was shown that by doing this we find modified riccati equations for the LQG problem so when we specialize this scenario above to the linear state dynamics linear up servation Milan quadratic control we find something that's close to LQG but not quite the similar idea was proposed in reference to four state dependent costs it was proposed to add additional factors in the factor graph related to the state cost with a regularizer λ and this λ could be tuned to obtain desired behavior we are now prepared to discuss the relation between active and France and Al Kochi recall that an active inference we're always trying to minimize free energy and this leads us to our main idea similar to the previous papers we will consider the future controls as hidden and there are states to be inferred secondly we will merge the control cost and the free energy objective different from before but without any loss of generality we will use a rolling horizon where at time t time little T we will have a horizon from the current time up to capital T time steps ahead so now we introduce our notation $\bar{y}$ are the observations up to time little T $\bar{y}$ over bar are the observations over a horizon of size capital T so from time T plus 1 to time T plus capital T similarly the states under bar $\bar{X}$ under bar represents the states up to time little T $\bar{x}$ over bar represents the future states $\bar{u}$ under bar represents the controls up to time t minus 1 $\bar{u}$ over bar represents the future controls including the control u_T that we want to infer we will then define a new generative model at time t denoted by P_T this model depends on all the observations all the hidden states all the future controls by this condition on the past controls

we factorize this distribution into three parts a normalization constant the system model which was the old generative model from early on in this presentation in which everything was conditioned on all of the controls times a goal prior and this goal prior depends on the few sure states the future observations and the future controls note that some variables such as X_T appear in the goal prior and in the system model this is fine because the normalization constant Z will make sure that the entire joint distribution is well-defined and normalized let us look at the shape of the proposed goal prior similar to the previous papers we consider to be in exponential minus λ times the cost over the horizon this cost as you may recall is quadratic in state and in the control the parameter λ will create different behaviors if λ is very large this means that our goal prior is very concentrated around zero zero and the control and zero and the state if λ is very small meaning close to zero this means that the goal prior is very flat in this state and control space with this generative model we can now define a free energy objective at time T this free energy objective involves a Kullback-Leibler divergence where P_T is a generative model which is given and Q is a distribution which we would like to determine by plugging in the definition of the Kullback-Leibler divergence as well as a definition of the generative model we can break down this free energy into two parts one part V_T which depends on the past and one part G_T which depends on the future the part G_T also contains the gold priors which relate to the future we show in the paper that the first term is minimized by a common filter and the second term is minimized by message passing backward in the factor graph the beliefs can then be found by running message passing on the corresponding factor graph with the gold priors and these beliefs found by message passing will then also minimize the free energy objective the principle outline here is called generalized free energy there is also a concept called expected free energy but this is different and will lead to different results the entire factor graph is shown here in green again we have the past and right an orange we have the future and what connects them is the state at time T the factor graph corresponding to the past is exactly the same as it was before and it is conditioned on all past controls the fact the factor F has as variables Y and $\bar{X}$ so there are the past observations in the past states the graph under riot corresponds to the future we see that there are these orange vertices in the graph and they correspond to the factorized gold prior we call that a gold prior is the exponential of a sum so this becomes a product of the Exponential's for each future state we have a gold prior for each future control we have a gold prior we also have gold price for future observations but these are set to uniform distributions we can now run message passing on this factor graph the messages on the factor graph are

visualized here oddish focus on some of these messages the message a from XT going from left to right is the Kalman filter output after the update so this is just a standard common filter running from left to right the message is B and C regarding future states they are backward messages in the factor graph which accounts for future goals this depends on the gold priors and future times now the message is e and D the message e dowered accounts for the gold prior of the control at time T and the message D upward accounts for the output of the Kalman filter a and the future goals C and B we can then multiply the messages E and D and this provides us the belief of the control at time T which we are interested in we can uh apply the mode of this belief to the system and shift our time horizon with one step and apply the same principle again the question now arises what can we prove about the relationship between active inference as proposed in the previous slide and LQG in the paper we prove the following two statements under a deterministic generative model active inference solves the stochastic optimal control problem so here the state dynamics and the observation model are fully deterministic secondly even when the generative model is not deterministic but λ becomes very small but positive then also active inference solves the underlying stochastic optimal control problem to make this specific for LQG we find the following expressions for the controller with the gain of the controller as well as the additional matrices that are also used in the riccati equations different from the standard riccati equations we have λ and many of these expressions but again it is possible to show that when either condition one or two holds then the λ cancels out in all these expressions and we recover exactly the lqg riccati equations so it is easy to verify that the under condition one or two the controller becomes exactly the lqg controller let us show some results to verify our claims this slide shows the performance of the LQG an active inference controller for a time window of 10 steps each of the four figures as on the x-axis time and under y-axis different quantities of interest on the bottom left we have the control magnitude which we would like to be small we have the state magnitude which we would also like to be small on the bottom right we have the accumulated control cost and on the top right we have a measure of the free energy in each of the figures we show three curves the lqg is in blue in red is the active inference controller with λ equal to one and in green is the active inference controller with λ equal to 0.1 we observe data LQG controller flies large controls in the beginning to bring the magnitude of the state close to zero and then apply small Corrections in contrast to active inference controller with λ equal to one is very sluggish and applies only very small controls needed to slow decrease of the state magnitude as a function of time the active inference controller with a

small λ equal to 0.1 based very close to the LQG controller this can also be seen in the bottom right showing the accumulated cost where lqg and the active inference with λ equal to 0.1 achieve similar accumulated cost while the active inference controller with a large λ leads to a large accumulated cost so these results confirm that for small λ the active inference Controller recovers lqg so it is interesting to see that larger λ leads to less aggressive control the lower free energy with higher cost and this result is somewhat counterintuitive one would expect that a larger λ would lead to more aggressive control bringing the state and the control magnitude close to zero quickly recall that with a large λ we will have a very concentrated goal prior around the target value of zero the interpretation of this is as follows if your model tells you that you will anyway end up where you want so this means with large λ your model tells you that you will have a control and a state close to zero then there is no need to work hard to get there so there is no need to apply large controls to reach your goal because you think you will get there anyway this was a counterintuitive but interesting result this now leads us finally to our conclusions active inference is a flexible and general framework can be applied to stochastic control and it has classical LQG as a special case provided we defined a free energy objective appropriately active inference automatically deals for different kinds of uncertainty including in the generative model itself the variational formulation is also good because it allows us to include various kind of objectives and constraint also it does not presume any separation principle as an algae it naturally emerges over all active inference is a rich and emerging area with lots of open research problems for the control signal processing and statistics community with this I would like to thank you for your time and your attention and your interest in our research